$$\frac{\partial\phi}{\partial b} = -2X'Y + 2X'Xb = 0 \qquad (10.26)$$

$$X'Xb = X'Y \qquad (10.27)$$

when X has rank equal to *k,* the normal equation 10.27 will have a unique solution and least squares estimator *b* is equal to:

$$b = [X'X]^{-1}[X'Y] \qquad (10.28)$$

We have assumed b to be estimator for β and thus E(b) = β , therefore we can rewrite 10.28 as

$$b = [X'X]^{-1} X''[X\beta + u]$$
$$= [X'X]^{-1} X'X\beta + [X'X]^{-1} X'U$$
$$= \beta + [X^1X]^1 X^1U$$
$$\therefore E(b) = E(\beta) + E\left[[X^1X]^{-1} X^1 E(\mu)\right] = E(B) + [X'X].X'E(\mu)$$
$$= \beta$$

Variance of $b = \sigma^2 (X'X)^{-1}$

Notes

1) In this course our objective is simply to introduce the concepts. Those who plan to pursue the concepts at much more rigorous level can study our course on Econometric Methods (MEC) – included as optional course of M.A.(Economics) Programme.

2) The other ideas regarding coefficient of determination R^2 and adjusted R^2 remain the same as they were developed for two explainatory variable case.

Now we can safely turn to discussions about non-satisfaction or violation of assumptions.

10.6 THE PROBLEM OF MULTI-CO-LINEARITY

Many a times our X variables may be found to have some other linear relationships among themselves. This vitiates our classical regression model.

Let us illustrate it with help of our 2 explainatory variable model.

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + U_1$$

Let us give specific names to variables, say, X_1 is price of commodity Y and X_2 is family income. We expect β_1 to be negative and β_2 to be positive. Now we go one step further. Let Y be demand for milk, X_1 be price of milk and let the family wise demand for milk is being estimated for a family, which produces and sells milk! Clearly, larger the value of X_1 higher the magnitude of X_2 will be.

In such situations, the estimation of co-efficient will not be possible. Recall, we wanted variables X in our matrix equations to be linearly independent. If that conditions is not satisfied X matrix becomes singular, that is its determinant tends to vanish. Thus, there will be **no solution** to the normal equation **10.26 (or 10.27).**

However, if co-linearity not perfect, we can still get OLS estimates and they remain best linear unbiased estimates (BLUE) – though one or more partial regression co-efficient may turn out to be individually insignificant.

Not only this, the OLS estimates still retain property of minimum variance. Further, it is found that multi co-linearity is essentially a sample regression problem. The X variables may not be linearly related in population but some of our suppositions while drawing a sample may create a situation of multiple linear relations.

The practical consequences of multi-co-linearity. Gujarali, (D.N.) has listed the following consequences of multiplicity of linear relationships:

1) Large variances /SEs of OLS estimates
2) Wider confidence intervals
3) Insignificant 't' ratios for β parameters
4) A high R^2 despite few significant t values
5) Instability of OLS estimators: The estimators and their standard errors (SEs) become very sensitive to small changes in data.
6) Sometimes, even signs of some of the regressions may turn out to be theoretically unacceptable like a rise in income having negative impact on demand for milk.
7) When many regressions have insignificant coefficients, their individual contributions to the explained sum of squares cannot be assessed properly.

The multi-co-linearity can be detected by:

1) high R^2 but few significant 't' ratios,
2) high pair wise correlation between explanatory variables. One can try partial correlations, subsidiary or auxiliary regressions as well. But each such technique increases burden of calculations.

Check Your Progress 2

1) When do you decide to drop a variable from the regression equation? Why?

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2) When do you include more variable(s) into your model? Why?

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3) "Inclusion of more variables always increases R^2 the goodness of fit. So to make a regression model 'good' what we need to do is simply increase the number of explanatory variables". Do you agree/disagree with this statement? Give reasons.

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4) What is multi-co-linearity? What are its consequences?

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10.7 PROBLEM OF HETERO-SCEDASTICITY

The Classical Linear Regression Model has a significant underlying assumption in homo scedasticity, that is, all the error terms are identically distributed with mean equal to zero and standard deviation equal to σ (or variance equal to σ2). σ2 What happens when this second part of assumption regarding distribution of variance does not hold? As a result, in symbolic terms $E(u_i)^2 = \sigma_i^2$, that is, if the expectation of squared errors is no longer equal to σ^2 — each error term has its own σ^2, or variance varies from observation to observation.

It has been observed that usually time series data does not suffer from this problem of hetero scedasticity but in cross-section data, the problem may assume serious dimensions. **The consequences of hetero scedasticity:** If the assumption of homoscedasticity does not hold, we observe the following impact on OLS estimators.

1) They are still linear

2) They are still unbiased

3) But they no longer have minimum variance – that is we cannot call them BLUE: the Best Linear Unbiased Estimators. In fact, this point is relevant both for small as well as large samples.

4) Reason for this problem hinted at in (3) above is that generally, OLS estimators have some bias built into their formulae. We try to rectify that making use of degrees of freedom.

   For instance $\hat{c}^2$, (the estimator for true population σ^2) given by $\sum e_1^2/df$ no longer remains unbiased. And this very $\hat{c}^2$ enters into calculation of standard errors of OLS estimates.

5) Since, estimates of standard errors are themselves no longer reliable, we may end up drawing wrong conclusions using conventional reasoning based on procedures for testing the hypothesis.

How to detect Hetero scedasticity

In applied regression analysis, plotting the residual terms can give us important clues about whether or not one or more assumptions underlying our regression model hold. The pattern exhibited by e_i^2 plotted against the concerned variable can provide important clue. If no pattern is detected – homoscedasticity holds or hetero scedasticity is absent. On the other hand, if errors form a pattern with variable — expanding, increasing linear or changing is some non-linear manner thus hetero scedasticity is certainly present.

Some tests have been designed to detect presence of Hetero scedasticity, using various statistical techniques. Prominent ones are: Park Test, Glejser Test, Whites General Test, Spearman's Rank correlation Test, Goldfield - Quadnt Test etc. But in this unit, the limitation of space does not permit us to go into their details. We are forced to refer the learners again the course on Econometric Method for details in this regard.

How to tackle the hetero scedasticity?

Our ability to tackle the problem will depend upon the assumptions. We can really make about error variance. Thus, the following situations may emerge

i) When σ_1^2 is known

Here the CLRM

$Y_i = \beta_0 + \beta_1 X_1 + u_i$ can be transformed, dividing each value by corresponding σ_1 thus,

$$\frac{Y_i}{\sigma_1} = \beta_0\left(\frac{1}{\sigma_1}\right) + \beta_1\left(\frac{X_i}{\sigma_i}\right) + \frac{U_i}{\sigma_i}$$

This effectively transforms error terms to $U_i = \frac{U_i}{\sigma_1}$ which is homo-scedastic and therefore, the OLS estimators will be free of disability caused by hetero scedasticity. The estimates of β_0 and β_1 in this situation are called **Weighted Least Squares Estimators (WLSEs)**.

ii) **When σ^2 is unknown:** we make some further assumptions about error variance:

iia) Error variance proportional to the X_i s. Here, the Square Root transformation is enough. We divide on both sides by $\sqrt{X_i}$. Thus, our regression line looks like:

$$\frac{Y_i}{\sqrt{X_i}} = \frac{\beta_0}{\sqrt{X_i}} + \beta_1 \frac{X_i}{\sqrt{X_i}} + \frac{U_i}{\sqrt{X_i}}$$

$$= \beta_0 \frac{1}{\sqrt{X_i}} + \beta_1 \sqrt{X_i} + U_i$$

Here $U_i = \frac{U_i}{\sqrt{X_i}}$ and this is sufficient to address the problem.

iib) Error Variance proportional to X_1^2. Here, instead of division by $\sqrt{X_i}$, we divide by X_i on both the sides and estimate

$$\frac{Y_i}{X_i} = \beta_0 \frac{1}{X_i} + \beta_1 + \frac{U_i}{X_i}$$
$$= \beta_0 \frac{1}{X_i} + \beta_1 + U_i$$

The error term will be $U_i = \frac{U_i}{X_i}$ and this will be free of hetero scedasticity, facilitating use of CLS techniques.

iii) ***Respecification of Model*:** Assigning a different functional form to the model, in place of speculating about the nature of variance may be found expedient. We can estimate this model:

$$\ln Y_i = \beta_0 + \beta_1 \ln X_i + U_i$$

This loglinear model is usually adequate to address our concerns.

10.8 PROBLEM OF AUTOCORRELATION

The classical regression model also assumes that disturbance terms U_is do not have any serial correlation. But, in many situations this assumption may not hold. The consequences of the presence of serial or auto correlation are similar to those of hetero scedasticity: the OLS are no longer BLUE.

Symbolically **no** autocorrelation means $E(U_i\ U_j) = 0$ when $i \neq j$.
Autocorrelation can arise in economic data on account of many factors:

i) there may be various reasons which cause cyclical up and down swings in economic time series. These tendencies continue till something happens which reverse them. This is called **inertia**.

ii) Misspecification of model in the form of under specification can be a cause of autocorrelation: you have fewer X variables in the model, leaving out rather large systematic components to be clubbed with errors.

iii) Cob-web phenomenon is another factor which may create this problem in certain types of economic time series (especially agricultural output etc.).

iv) ***Polishing of data*** – like adding monthly data to make quarterly or quarterly to **make** half yearly series etc. can also be responsible for the autocorrelation – as the averaging involved dampens the fluctuations of the original data.

The consequences of autocorrelation are not different from those of hetero scedasticity listed in 10.7 above. Here too OLS estimators are biased or are not BLUE, t & F tests are no longer reliable. Therefore, computed value of R^2 is not reliable estimate of true goodness of fit.

There are many tests for detecting autocorrelation – varying from visual inspection of error plots, the Runs Test, Swed-Eisenhart critical runs test. But most commonly used in Durbin-Watson d test defined as

$$d = \frac{\sum_{t=2}^{n}(e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2}$$

However, again, we are holding back information on practical detections and avoidance of problem of autocorrelation for the reasons of limitation of space here.

10.9 THE MAXIMUM LIKELIHOOD ESTIMATIONS

Sometimes another class of estimators is used in place of OLS. This class of estimators is called ***maximum likelihood estimators*** (MLEs). The MLEs possess some stronger theoretical properties. But it also requires a stronger assumption about distribution of error terms. Moreover, when errors follow normal distribution, we find that OLS and MLE methods give identical estimates of β parameters, both in simple and in multiple regressions. However, MLE estimate of σ^2 is biased. Hence, if one uses assumption of normal distribution of U_is and persists with OLS, one does not miss out on any thing that may be advantageous in MLE.

The Method: Simple Regression Illustration

In the model $Y_1 = \beta_0 + \beta_1 X_1 + U_i$, the Y_1 is normally and independently distributed with means $\beta_0 + \beta_1 X_i$ and variance σ^2. Therefore, the joint probability density function of

$Y_1, Y_2, Y_3 \ldots\ldots\ldots\ldots\ldots\ldots\ldots Y_n$ with above mean and variance will be

$$f\left(Y_1 \ldots\ldots\ldots\ldots\ldots Y_n / \beta_0 + \beta_1 X_i, \sigma^2\right) \qquad \text{ML - 1}$$

But given the independence of Y_s, this function can be written as product of 'n' individual density functions, or

$$f\left(Y_1 \ldots\ldots\ldots\ldots\ldots Y_n / \beta_0 + \beta_1 X_i + U_i, \sigma^2\right) \qquad \text{ML-2}$$

$$= \pi\left[Y_i / \beta_0 + \beta_1 X_i + U_i, \sigma^2\right]$$

$$\text{where} \quad f(Y_i) = \frac{1}{\sigma\sqrt{2\pi}} . e^{\left[-Y_2 \frac{(Y_i - \beta_0 - \beta_1 X_i)^2}{\sigma^2}\right]} \qquad \text{ML-3}$$

Putting this value for each Y_i, in ML-2. We get the likelihood function (LF):

$$LF(\beta_0, \beta_1, \sigma^2) = \frac{1}{\sigma^n \left(\sqrt{2\pi}\right)^n} . e^{\left[-\frac{1}{2}\sum \frac{(Y_i - \beta_0 - \beta_1 - U_i)^2}{\sigma^2}\right]} \qquad \text{ML - 4}$$

The method of maximum likelihood estimation is nothing but maximization of the (LF) given in ML-4 above. We can differentiate logarithms of (LF) with respect to β_0, β_1 and σ^2 and equate the respective partials to zero to get the requisite estimation relations. So as

$$\ln(LF) = n \Big/ n\sigma^2 - \frac{n}{2}\ln(2\pi) - \frac{1}{2}\sum \frac{(Y_i - \beta_0 - \beta_1 X_1 - U_i)^2}{\sigma^2} \qquad \text{ML - 5}$$

therefore

$$\frac{\partial \ln(\text{LF})}{\partial \beta_0} = \frac{-1}{\sigma^2}\sum (Y_1 - \beta_0 - \beta_1 X_i)(-1) = 0 \qquad \text{ML - 6}$$

$$\frac{\partial \ln(\text{LF})}{\partial \beta_1} = \frac{-1}{\sigma^2}\sum (Y_1 - \beta_0 - \beta_1 X_1)(-X_i) = 0 \qquad \text{ML - 7}$$

$$\frac{\partial \ln(\text{LF})}{\partial c^2} = \frac{-n}{2c^2} + \frac{1}{2c^4}\sum (Y_1 - \beta_0 - \beta_1 X_i)^2 = 0 \qquad \text{ML - 8}$$

Simplification of ML-6 and ML-7 give us

$$\Sigma Y_i = n\beta_0 + \beta_1 \Sigma X_i \qquad \text{ML-8a}$$

$$\Sigma X_i Y_i = \beta_0 \Sigma X_i + \beta_1 \Sigma X_i^2 \qquad \text{ML-9}$$

Which are same as OLS normal equations.

Substituting values of ML (=OLS) estimates obtained by simultaneously solving ML-8 and ML-9 into ML-7 gives us

$$\sigma^2 = \frac{1}{n}\sum (Y_i - \beta_0 - \beta_1 X_i)^2$$

$$= \frac{1}{n}\sum U_i^2 \qquad \text{ML - 10}$$

Expectation of $\tilde{\sigma}^2, E(\tilde{\sigma}^2) = \frac{1}{n}E(\Sigma U_i^2)$

$$= \left(\frac{n-2}{n}\right)\sigma^2$$

$$\sigma^2 - \frac{2}{n}\sigma^2$$

or $\tilde{c}^2$ is biased downwards.

Multiple Regression: An example: The following regression were run on SPSS. Edited summary of results is as follows:

India's Imports, Exports and Foreign Investment Inflows

All figures are in Millions of US dollars

Year	var00001	var00002	var00003
1991-1992	19411.00	17865.00	133.00
1992-1993	21882.00	18537.00	559.00
1993-1994	23306.00	22238.00	4153.00
1994-1995	28654.00	26330.00	5138.00
1995-1996	36678.00	31797.00	4892.00
1996-1997	39133.00	33470.00	6133.00
1997-1998	41484.00	35006.00	5385.00
1998-1999	42389.00	33218.00	2401.00
1999-2000	49671.00	36822.00	5181.00
2000-2001	50536.00	44560.00	5862.00
2001-2002	51413.00	43827.00	6693.00
2002-2003	61412.00	52719.00	4555.00

var00001=Imports

var00002=Exports

var00003=Foreign Investment Inflow

Regression of Imports on Exports and Foreign Investment Inflow

R	R^2	$\bar{R}^2$	Standard error of Estimate
0.983	0.966	0.958	2726.228

Coefficients

	B	Standard Error	t
Constant	-1655.737	2658.578	-0.623
var00002	1.263	0.104	12.192
var00003	-0.288	0.522	-0.552

Coefficients Correlation

Correlation		Var00003	Var00002
	Var00003	1.0000	-0.670
	Var00002	-0.670	1.0000
Covariance	Var00003	0.272	-3.625E-02
	Var00002	-3.625E-02	1.073E-02

ANOVA

Sum of Sqrs.	*df*	Mean sqrs.	F
Regression1.9E+09	2	9.4E+08	127.097
Residual 6.7E+07	9	7432320	
Total 2.0E+09	11		

Regression of Foreign Investment Inflow on Exports and Imports

R	R^2	$\bar{R}^{-2}$	Standard error of Estimate
0.684	0.468	0.349	1712.334

Coefficients

	B	Standard Error	t
Constant	-321.545	1702.075	-0.189
var00001	-0.114	0.206	-0.552
var00002	0.272	0.257	1.060

Coefficients Correlation

Correlation		**Var00002**	**Var00001**
	Var00002	1.0000	-0.982
	Var00001	-0.982	1.0000
Covariance	Var00002	6.590E-02	-5.92E-02
	Var00001	-5.192E-02	4.24E-02

ANOVA

	Sum of Sqrs.	*df*	Mean sqrs.	F
Regression	2.3E+07	2	1.2E+07	3.952
Residual	2.6E+07	9	2932089	
Total	5.0E+07	11		

The above regressions are based on certain 'a prior' expectations. We expect that imports into India during the period 1991-92 to 2002-03 depend upon exports from India and foreign investment inflows into the country. The idea is that exports pay for imports and foreign investment inflow 'facilitates' the country to import more. Our results confirm our theoretic expectations. Regression on imports on exports and investment inflows shows that R^2 = 0.966 and $\bar{R}^2 = 0.958$. This shows that model has high explanatory power. It explains over 95 per cent of the variation. Typical computer output gives information on coefficients, correlation and co-variances, co-linearity diagnostics, residual analysis of variance etc. We have run another regression too – the results of which are reported above. This is regression of foreign investment inflows on exports and imports. Theoretic expectation was that investment inflows are determined by magnitudes of exports and imports. However, this model gives rather disappointing results. $R^2 = 0.468$ and $\overline{R^2} = 0.349$ only. In other words, our model explains less than 35 per cent of the variations. It is not desirable to persist with it. The same is reflected in low values of β_1 & β_2 parameters and their high standard errors.

Check Your Progress 3

1) What is hetero-scedasticity? What are its consequences?

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2) What is auto-correlation? When does it arise? What are its consequences?

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3) What are maximum likelihood estimations? Why do we persist with least squares estimators most of the time.

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10.10 LET US SUM UP

We began with extension of the simple linear regression model to incorporate one more explanatory variable. Our next step was to interpret the coefficients of partial regression. Afterwards, we tried to design statistical touch stone for inclusion of more variables into the model. This also gave us some guidelines to drop the 'undesirable' variable as well. We then moved on to more tedious and mathematically more demanding extension to 'n' variable model. We then attempted to analyse the effects of multi-co-linearity, hetero scedasticity and auto-correlation respectively. We also made comments on techniques of identification of these problems and some strategies to get rid of the problems as well. Further, we discussed the class of estimators called maximum likelihood estimators (MLEs). We made comparison between MLEs and Least square estimators as well. Finally, we ran some regressions on SPSS package between India's imports, exports and foreign investment inflows to illustrate some of the points, which arose in course of our discussions. However, we must again draw the attention of learners to the fact that limitations of space did not permit us to go in for an exhaustive discussion of the concepts touched upon. Those who want to have detailed knowledge about the issues and concepts involved in multiple regression models are advised to go through our optional course MECE-001 Econometric Methods.

10.11 KEY WORDS

Multiple Regressions : Regression of one dependent variable on more than one independent variables.

Partial regression Coefficients : Coefficients of individual predictors in multiple regressions.

Coefficient of multiple determination — R^2 : Ratio of variation explained or accounted for by the regression model to the total variation.

Adjusted coefficient of multiple determination — $\overline{R}^2$: It is coefficient of determination adjusted for the loss of degrees of freedom. It helps us against increasing $\overline{R}^2$ by incorporating unnecessary explanatory variables.

Multi-co-linearity : Existence of linear relationships between explanatory variables in addition to one between dependent variable and them.

Hetero scedasticity : Differences between variances of the error terms. This problem is more serious in cross-sectional data.

Auto-co-relation : Correlations between the errors and their previous values. This problem is encountered very often when dealing with time series data.

10.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

1) Read Section 10.2
2) Read Section 10.2
3) Read Section 10.2

4) Read Section 10.3
5) Read Section 10.4
6) Read Section 10.4
7) Read Section 10.4
8) Read Section 10.6
9) Read Section 10.7
10) Read Section 10.8
11) Read Section 10.9